KQML · KIF · ONTOLOGY

AGENT DIALOGUE: ALICE AND BOB



Alice · Procurement agent



Bob · Warehouse agent

Scenario: Alice needs to procure 50-inch televisions. She asks Bob to check the available stock and determine how many HDMI slots each television has.

STEPS 1–2 · AVAILABLE STOCK

1 · ALICE → BOB

Alice sends Bob a KQML ask-one with a KIF query: how many TV50 units are in stock?

```
(ask-one
  :sender Alice
  :receiver Bob
  :language KIF
  :ontology warehouse-inventory
  :content (available-stock TV50 ?quantity))
```

2 · BOB → ALICE

Assuming the warehouse has 25 televisions, Bob replies with a tell.

```
(tell
  :sender Bob
  :receiver Alice
  :language KIF
  :ontology warehouse-inventory
  :content (available-stock TV50 25))
```

STEPS 3–4 · HDMI SLOTS

3 · ALICE → BOB

Alice sends a second query: how many HDMI slots does a TV50 have?

```
(ask-one
  :sender Alice
  :receiver Bob
  :language KIF
  :ontology warehouse-inventory
  :content (hdmi-slots TV50 ?number-of-slots))
```

4 · BOB → ALICE

Assuming each television has 3 HDMI slots, Bob responds with a tell.

```
(tell
  :sender Bob
  :receiver Alice
  :language KIF
  :ontology warehouse-inventory
  :content (hdmi-slots TV50 3))
```

SUMMARY AND KEY CONCEPTS

KQML

The communication layer: who sends, who receives, and the type of act.

KIF

The content: the knowledge inside the message.

ONTOLOGY

The shared vocabulary for TV50, available-stock and hdmi-slots.

#	Direction	KQML	KIF content
1	Alice → Bob	ask-one	(available-stock TV50 ?quantity)
2	Bob → Alice	tell	(available-stock TV50 25)
3	Alice → Bob	ask-one	(hdmi-slots TV50 ?number-of-slots)
4	Bob → Alice	tell	(hdmi-slots TV50 3)